CSC1042 – Computer Networks 2

Lab 2 - Connect devices and configure their IP addresses

Addressing Table

Device	Interface	IP Address	Connects To
Router 3	FastEthernet1/0	192.168.30.1/24	FastEthernet0/0
	FastEthernet0/0	192.168.5.1/24	Fast Ethernet0/1
Router 1	Console	N/A	RS 232
	FastEthernet1/0	10.0.0.1/8	FastEthernet0
	Serial 2/0	172.31.0.1/16	Serial 3/0
Router 2	Serial 3/0	172.31.0.2/16	Serial 2/0
	FastEthernet1/0	172.16.0.1/16	FastEtherent0/1
	FastEthernet0/0	192.168.30.2/24	FastEthernet1/0
Home PC	FastEthernet0	192.168.5.100/24	FastEthernet0/2
Family PC	FastEthernet0	192.168.5.102/24	FastEthernet0/3
Web server	FastEthernet0	10.0.0.254/8	FastEthernet0/1
Configuration PC	RS 232	N/A	Console

Background

When working in Packet Tracer, you should know how to select the appropriate cable and how to properly connect and configure devices. This activity will examine device configurations in Packet Tracer, selecting the proper cable based on the configuration, connecting the devices, and assigning them IP addresses.

Instructions

Open the **Lab2.pka** file. When the **User Profile** window appears, enter your **full name** and **email address** exactly as registered on Loop. Failure to provide this information will result in a **0** grade for this lab assignment.

Part 1: Connect Router 1

Step 1: Connect Router 1 to Router 2.

Choose the correct cable to connect **Router 1 Serial2/0** to **Router2 Serial3/0**. Use one of the available **Serial** cables.

If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect Router 1 to Web server.

Choose the correct cable to connect **Router 1 FastEthernet1/0** to **Web server FastEthernet0**. Routers and computers traditionally use the same wires to transmit (1 and 2) and receive (3 and 6). The correct cable to choose consists of these crossed wires. Although many NICs can now autosense which pair is used to transmit and receive, **Router 1** and **Web server** do not have autosensing NICs.

If you attached the correct cable, the link lights on the cable turn green.

Step 3: Connect Router 1 to the Configuration PC.

Choose the correct cable to connect **Router 1 Console** to **Configuration PC RS 232**. This cable does not provide network access to **Configuration PC** but allows you to configure **Router 1** through its terminal.

If you attached the correct cable, the link lights on the cable turn black.

Part 3: Connect Remaining Devices

Step 1: Connect Router 2 to Switch 1.

Choose the correct cable to connect **Router 2 FastEthernet1/0** to **Switch FastEthernet0/1**.

If you attached the correct cable, the link lights on the cable turn green. Allow a few seconds for the light to transition from amber to green.

Step 2: Connect Router 2 to Router 3.

Choose the correct cable to connect **Router 2 FastEthernet0/0** to **Router 3 FastEthernet1/0**. If you attached the correct cable, the link lights on the cable will turn green.

Step 3: Connect Router 3 to Switch 2.

Choose the correct cable to connect **Router 3 FastEthernet0/0** to **Switch 2 FastEthernet0/1**. If you attached the correct cable, the link lights on the cable turn green.

Step 4: Connect remaining devices to Switch 2.

Choose the correct cable to connect **Switch 2 FastEthernet0/2** to **Home PC FastEthernet0**. If you attached the correct cable, the link lights on the cable turn green.

Choose the correct cable to connect **Switch 2 FastEthernet0/3** to **Family PC FastEthernet0**. If you attached the correct cable, the link lights on the cable turn green.

Part 4: IP addresses assignment

Step 1: Assign IP address to the interfaces of Router 1 as shown in the Addressing Table.

a. Click on **Router 1** and select the **Config Tab** from the top menu.

b. Configure **Router 1's interfaces** based on the information in the Addressing Table.

Step 2: Repeat the same process for Routers 2 and 3

Step 3: Assign IP address to Web server

a. Click on **Web Server** and select the **Desktop Tab** from the top menu.

b. Click on the **IP configuration** application.

c. Configure **Web server** IP address based on the information in the Addressing Table. You should fill in the IP address, the subnet mask, and the default gateway

Step 4: Repeat the same process for Home PC and Family PC

Part 5: Verify Connections

Step 1: Test the connection between Family PC and Web Server.

a. Open the Desktop tab on the **Family PC** and select the command prompt application.

b. Ping **Web server's** IP address.

c. Open the **Web Browser** and type in the web address **www.example.com.** You might need to wait for 30 to 60 minutes before the web page is displayed.

Step 2: Open Router 1 from Configuration PC.

a. Open the **Terminal** application of **Configuration Terminal** and accept the default settings.

b. Press **Enter** to view the **Router 1** command prompt.

c. Type **show ip interface brief** to view interface statuses.